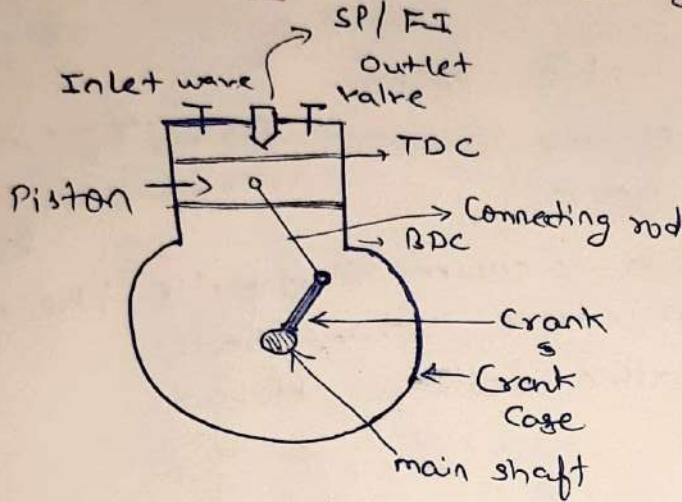


## Unit 2 I.C Engine & Pump

### \* IC Engine

SP  $\Rightarrow$  Spark plug  
FI  $\rightarrow$  Fuel injector

TDC  $\rightarrow$  Top dead center  
BDC  $\rightarrow$  Bottom dead center.



IC Engines are heat engines where combustion of fuel occurs inside the engine cylinder. They convert chemical energy into mechanical energy.

### classification-

- Based on fuel
  - $\rightarrow$  Petrol (spark ignition)
  - $\rightarrow$  Diesel (Compression ignition)
- Based on Number of strokes
  - $\rightarrow$  Two-stroke engine
  - $\rightarrow$  Four-stroke engine
- Based on Cooling method
  - $\rightarrow$  Air cooled
  - $\rightarrow$  Water-cooled
- Based on number of cylinder
  - $\rightarrow$  Single cylinder
  - $\rightarrow$  Multi cylinder.

### Main Components

- \* cylinder  $\rightarrow$  Chamber where fuel combustion takes place
- \* piston  $\rightarrow$  Moves up and down inside the cylinder.
- \* connecting Rod  $\rightarrow$  Connect piston to crankshaft.
- \* Crank shaft  $\rightarrow$  Converts reciprocating motion to rotary motion.
- \* Valves  $\rightarrow$  Control intake & exhaust of gases
- \* Spark plug  $\rightarrow$  (in petrol): ignites the air-fuel mixture
- \* fuel injector (in diesel) Inject fuel into combustion
- \* Cooling System  $\rightarrow$  Maintain temp
- \* Lubrication system  $\rightarrow$  Reduce friction.



## \* Four-Stroke I.C Engines

### ① Intake (Suction) Stroke -

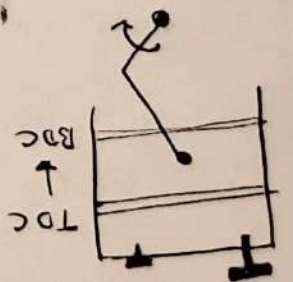
To draw the air-fuel mixture or air into cylinder.

Process - The intake valve opens

piston moves downward from TDC to

BDC

- This creates a vacuum that pulls the air or fuel mixture into cylinder.
- Exhaust valve remains closed.



### ② Compression stroke -

To compress the air-fuel mixture to increase temp.

Process - Both intake & exhaust valve are closed.

The piston moves upward from BDC to TDC

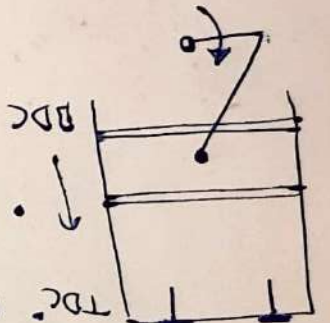
mixture is compressed in the

combustion chamber.

At the end of this stroke.

A spark plug ignites the mixture

Fuel is injected into hot compressed air.



### ③ Power stroke

To generate power from the combustion of fuel.

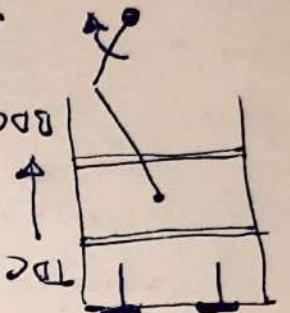
Process - Combustion occurs, rapidly increasing pressure

forces the piston down from TDC to BDC

This stroke produces useful work & rotates

the crankshaft

BDC. Both valve remain closed.





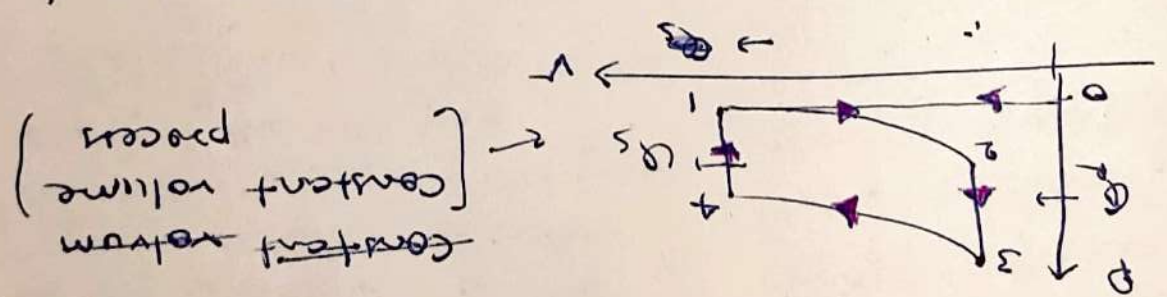
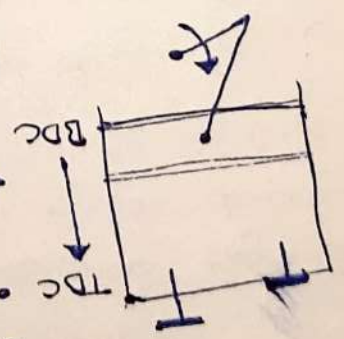
# ④ Exhaust stroke - can

To expel burnt gases from the combustion chamber.

Process -

- exhaust valve opens.
- piston moves up from BDC to TDC
- burnt gases are pushed out through the exhaust valve
- intake valve remains closed.

Note in 2 rotation crank → 1 power stroke.



• Difference between 4 stroke petrol engine & diesel engine

\* SI engine 4 stroke petrol engine

→ Ignition - spark (SI)

Fuel - petrol

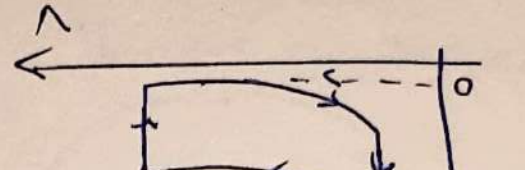
Efficiency - lower

weight - lighter

maintenance - easier

cheaper constant volume → 10 bar pressure

Application - Cars / bikes / Scooter



8 to 12 (Combustion ratio)

\* CI engine 4 stroke diesel engine

Compression (CI)

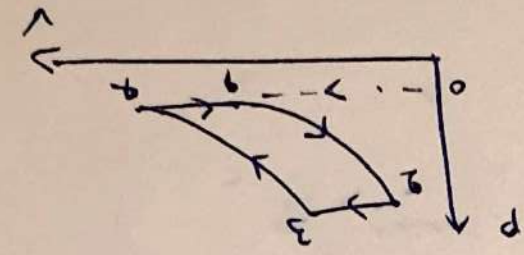
Higher

Heavier

Costly & complex.

constant pressure = 33 bar

Trucks, buses, tractors, heavy machinery.



14 to 25 (Combustion ratio)



## \* 2 Stroke I.C engine - (Working)

### Stroke 1

(Compression + Suction)

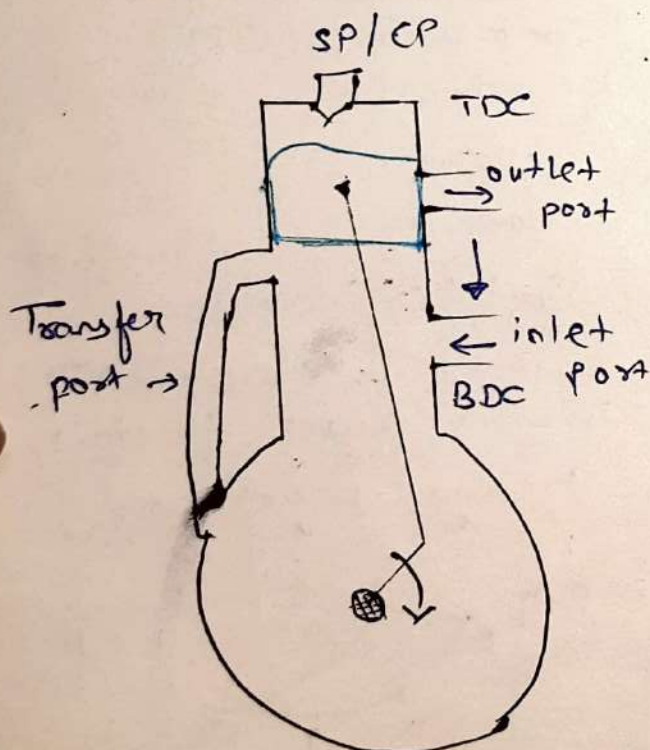
Piston moves upward from Bottom (BDC) to TDC

→ It compresses fuel mixture inside the combustion engine.

• At the same time, partial vacuum is created in the crankcase, drawing in a fresh air/fuel the inlet port.

→ Just before the piston reaches TDC

Spark plug or fuel injected into the compressed mixture.



### Stroke 2

(Power + Exhaust)

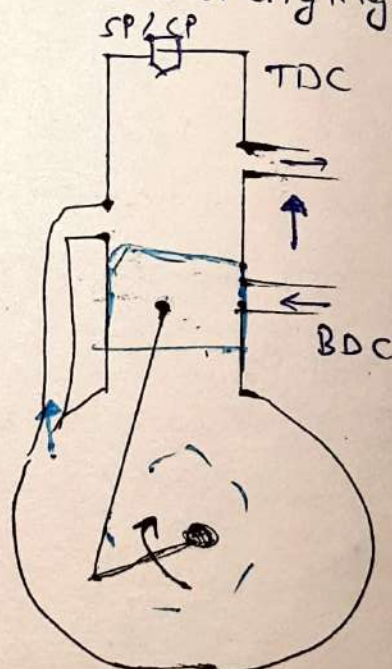
Due to combustion, high pressure forces the piston downward from BTDC to BDC.

→ As piston moves down, compressed the fresh charge in the crankcase.

→ First, exhaust port opens, allowing burnt gases to escape.

→ Transfer port opens & the fresh charge enters the cylinder, pushing out remaining exhaust gases.

(Scavenging).



partial compression process.



## main components

Cylinder, piston, spark plug / combustion plug,  
port  $\rightarrow$  inlet, outlet, transfer, crankcase

## Difference between 2 stroke & 4 stroke

### 2 stroke

- 1) 2 rotation of crank, 1 power stroke
- 2) Valves are used
- 3) Cam & follower mechanism is used
- 4) Size of fly wheel is more
- 5) Size of engine is more (weight)
- 6) Flat face head of piston is used.
- 7) Thermal efficiency is more
- 8) Cost is more
- 9) maintenance is more

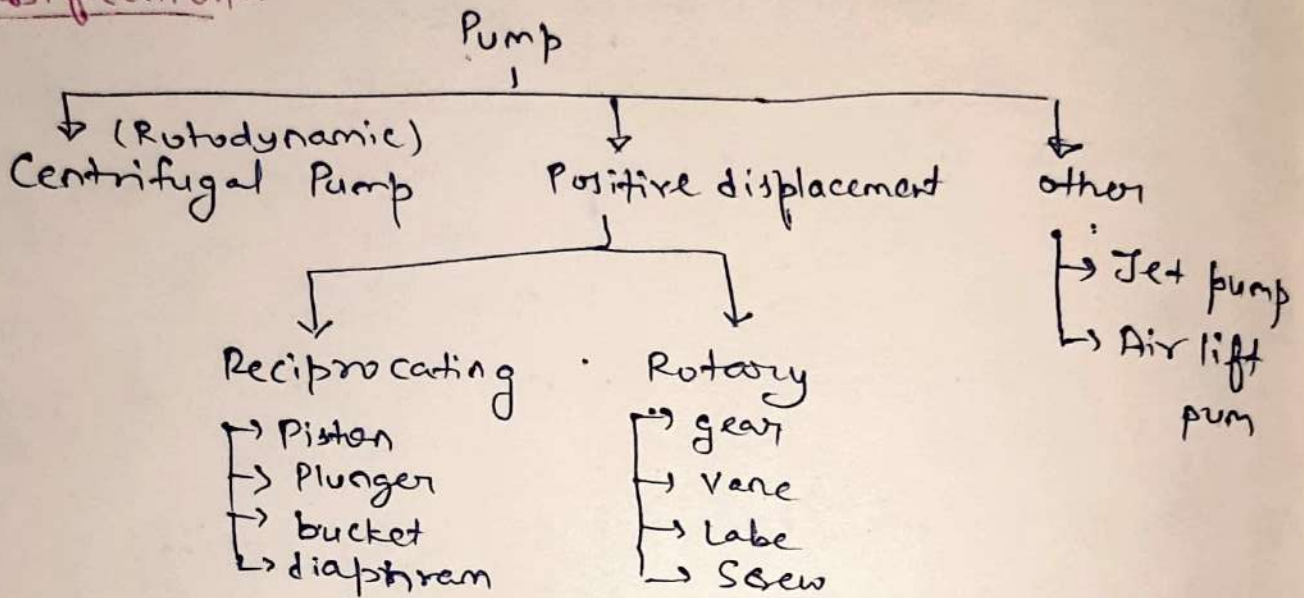
### 4 stroke

- 1 rotation of crank, 1 power stroke.
- Ports are used.
- Cam & follower mechanism is not used.
- Size of fly wheel is less
- Size of engine is less (weight)
- Aerofile head of piston is used.
- Thermal efficiency is less.
- Cost is less
- maintenance is less.

\* IP  $\Rightarrow$  Indicated power  
BP  $\rightarrow$  Brake power

\* Pump :- Pump is a device which is used to transfer the liquid from one place to another by increasing the head (Pre.) of liquid.

classification -

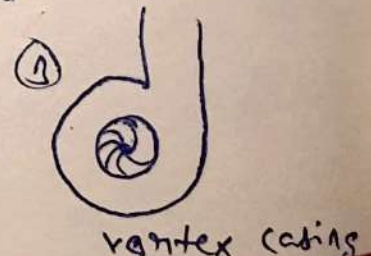
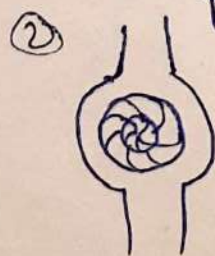
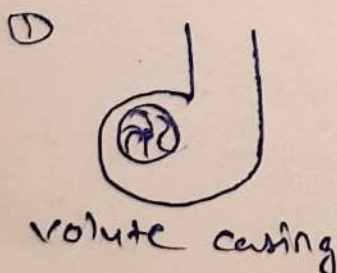
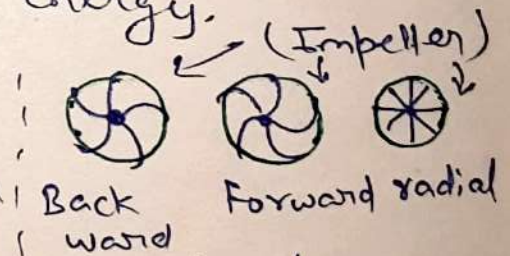
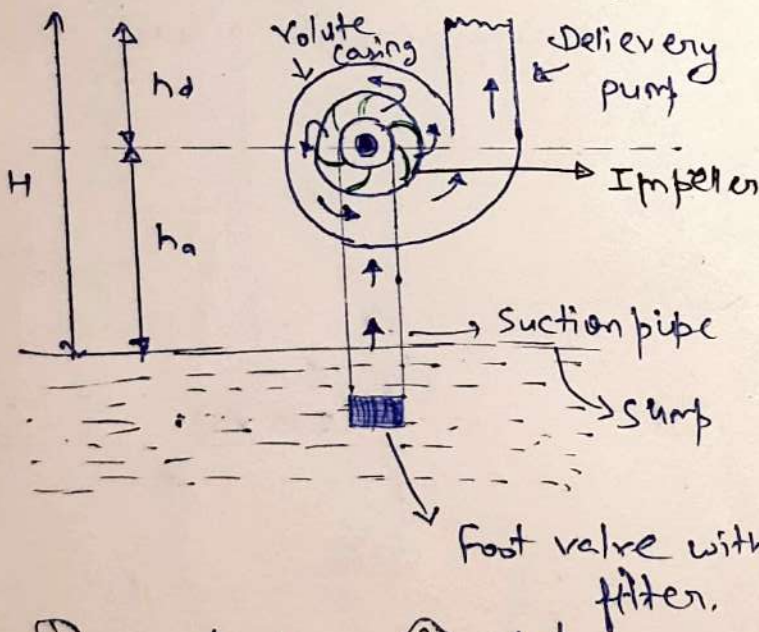


In Syllabus → Centrifugal pump.  
Reciprocating

### (1) Centrifugal Pump -

\* These pumps use centrifugal force generated by a rotating impeller to move fluid.

It convert mechanical energy into kinetic energy.



guide blades casing.

vortex casing



## Parts

Delivery pipe - Used to transfer liquid.

Impeller - Rotating part used to increase pressure of liquid.

Suction pipe - One end inside the liquid & another is connected with pipe pump.

Foot valve → One way valve. Allow water in upward direction.

Priming cup → work as coolant, To remove air bubble from casing.

Casing - Provide protection cover to the pump.

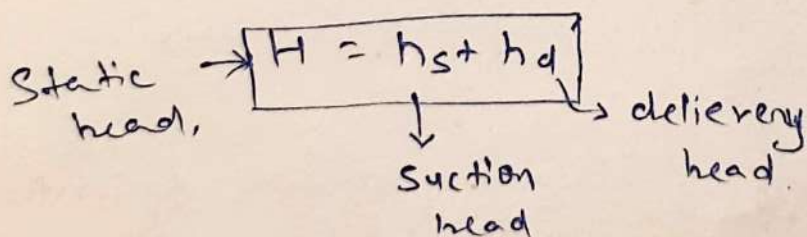
Filter or strainer → screen liquid.

Working - Fluid enters the pump impeller along or near to the rotating axis & is accelerated by the impeller.

Kinetic energy of the fluid is converted to pressure energy in the volute casing & the fluid is discharged at high pressure.

## Application

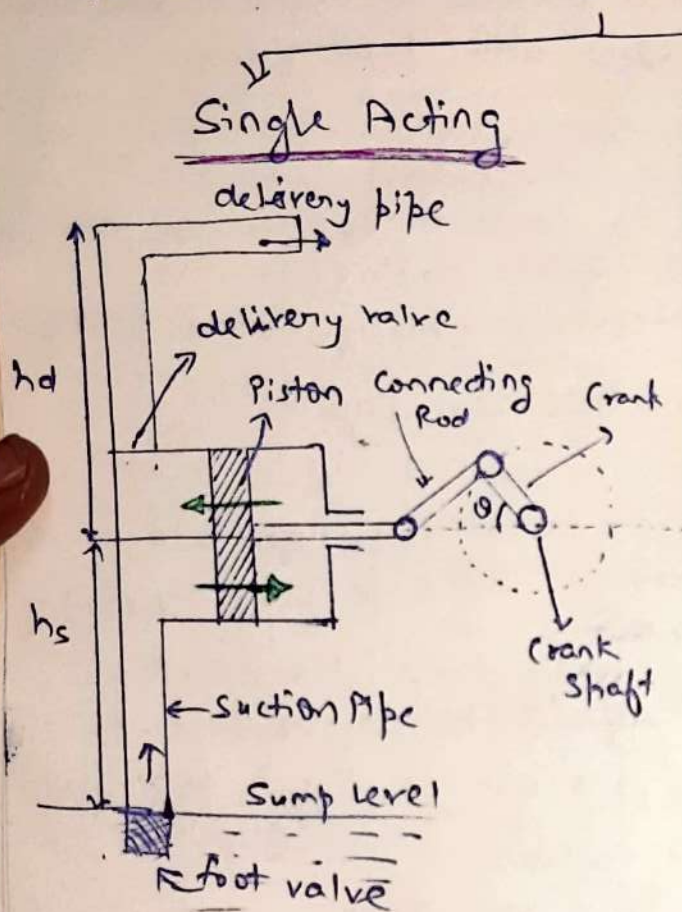
- Domestic water supply
- Irrigation system.
- Chemical processing plants
- Cooling water system.





## (a) Reciprocating Pump:-

Reciprocating Pump operates on the principle of a piston moving back & forth within a cylinder to draw & discharge fluid.

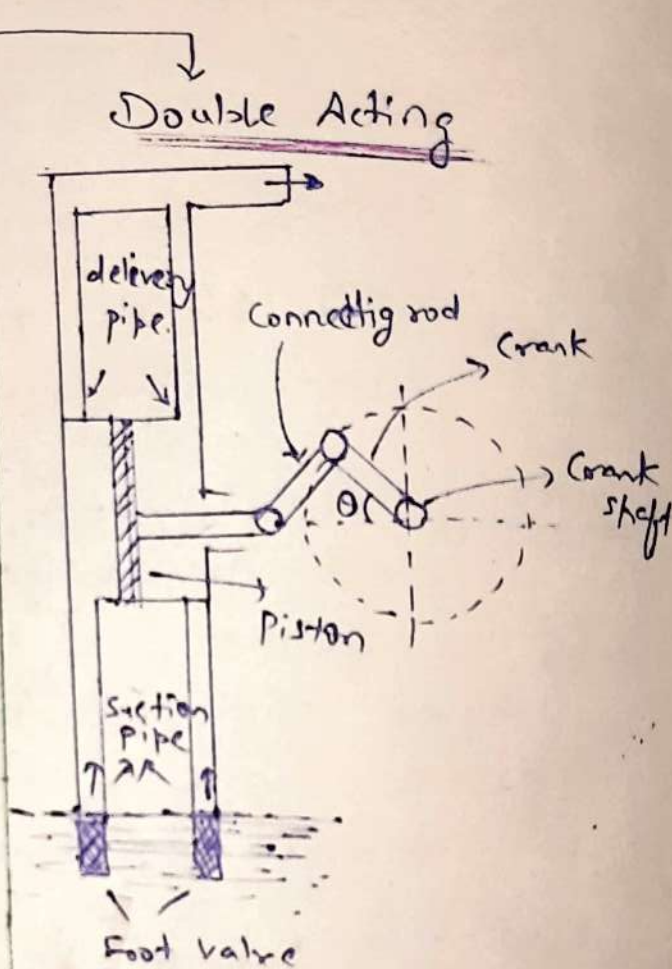


Working

- Only one suction pipe
- Only one delivery pipe
- Water acts on one side of Piston

• During one rotation of shaft, there is only one delivery Stroke

- Water pumped will be less
- Less power required
- Cost is less



Working

- Two suction pipes.
- Two delivery pipes.
- Water acts both side of piston.

- Two delivery strokes

- Double amount of water will be pumped.
- High power required
- Cost is more



## \* working (for both)

- Suction Stroke: - The piston moves away from the cylinder head, creating a vacuum that draws fluid into the cylinder through a Suction valve.
- Delivery stroke: - The piston moves towards the cylinder head, increasing pressure & pushing the fluid out through a delivery valve.
- Application - Used in hydraulic system
  - Oil drilling rigs
  - High-pressure cleaning systems.
  - Water supply to boilers.

Note Cavitation - Formation & rapid collapse of vapor bubbles in a liquid, due to pressure drop.

→ it can be removed by increasing the pressure at the pump inlet, reducing fluid temp & optimizing pump & system design.